TORUS BOUNTY · RUSH HOUR FOR ELECTRONS

# HUM

the grid, heard

The grid broadcasts its state sixty times a second, through the wire in the wall.

This is what it sounds like, and what a Torus Station should do about it.



SPIN · PULSE · LASSO

# THE EAR

a guitar pickup, a cable, a room




The Western Interconnection, measured from this room's wiring through a guitar pickup.

A pickup is a coil. Every wire in this room hums at sixty.

The hum's pitch is not the room's. It belongs to the whole Western Interconnection, Alberta to Baja. It sags when the grid is short.

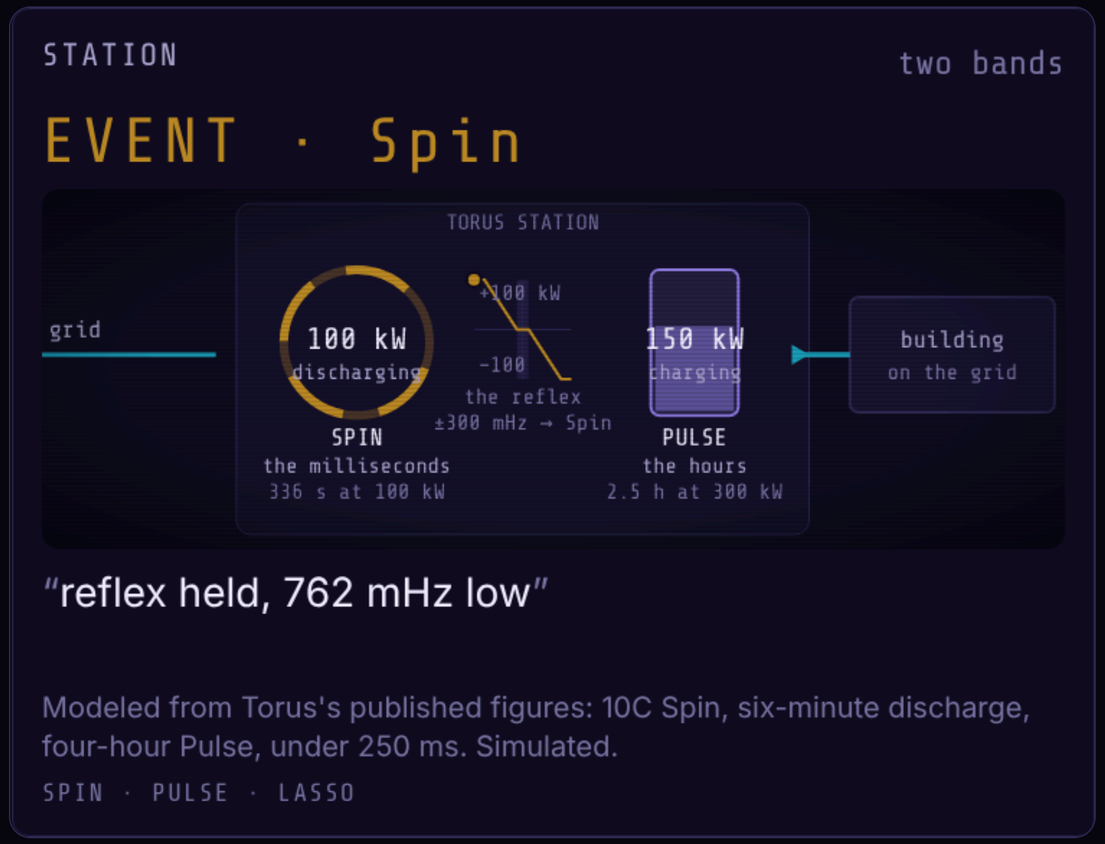
The grid is a note. Sixty hertz is a B, forty-nine cents flat. One cent is thirty-five millihertz, so the grid's whole quiet life is the tenths digit on a tuner.

Read to the millihertz, three independent ways that agree, with the DSP library behind my pedals.

The same code runs on a thirty-dollar chip inside a guitar pedal.

# TWO BANDS

a Station answers at two speeds




Spin takes the milliseconds. A plant trips, the frequency falls, Spin answers within a third of a second. The reflex is a compressor on the grid's pitch: the deadband is its threshold, the droop its ratio, ten C its ceiling.

Pulse takes the hours. The rush hour is on the page before it arrives; Pulse carries the building through it and refills in the clean hours. Price belongs here, in the hour's plan.

When the hum dies, the Station carries the building. The same ear knows first. Two-way, as you said.

On 9 August 2019 the British grid changed key, B to A♯, and a million people lost power at that A♯; it replays here, transposed and labeled. The Station is simulated from Torus's published figures and labeled so.

# THE PROGRAM THAT DECIDES

proven, not reviewed

### WHAT THE PROGRAM THAT DECIDES CAN AND CANNOT DO

PROVEN · MENTL CHECK

**CAN**

- ✓ act on each reading the loop hands it
- ✓ write a command for Spin and for Pulse
- ✓ keep its own state of charge and its timers, on lines it owns

**CANNOT**

- ⊘ reach the network while deciding
- ⊘ touch the filesystem while deciding
- ⊘ allocate memory mid-decision
- ⊘ ask Spin for more than 100 kW, or Pulse for more than 300 kW
- ⊘ accept a frequency outside 55 to 65 Hz as real

```
$ mentl why examples/grid/station.mn decide Network
the row of decide carries no Network
```

Not by policy: proven by the compiler through every call the program makes. Press A for the compiler's own words.

### THE SAME PROGRAM, ONE PRICE READ ADDED INSIDE THE REFLEX

REFUSED

```
fn decide(valid, hz, rocof, level, hour, rush, clean, out) -> Mode with !Alloc + !Network + !Filesystem = {
  ...
  let price = price_m("https://api.eia.gov/v2/electricity/rto/region-data/…")
  let pulse_ask = if rush == 1 { if price > 50000 { 300000 } else { 200000 } } else { … }
  ...
}
```

`decide` → `price_m` → `fetch_param` → `Network` the path the network takes into a function that swore it never would

```
effects: E_EffectMismatch error: effect row mismatch: !Alloc + !Network + !Filesystem vs Mutate + Memory + Trap + Network — at 107:1-228:2: declared as decide; Network enters at examples/grid/station-leak:176:5 through `price_m` → at examples/grid/station-leak:73:14 through `fetch_param` → declared as fetch_param, an op of Network declared at dsp/signal:66:1 at examples/grid/station-leak:107:1-228:2
effects: E_EffectUnhandled error: effect Network reaches the executable root with no absorbing handler — calling one of its ops would read garbage evidence at runtime; nothing executes unproven. No handler in scope handles Network — declare one and install it with ~> over the performing chain. declared as main; Network enters at examples/grid/station-leak:428:3 through `step` → at examples/grid/station-leak:409:16 through `decide` → at examples/grid/station-leak:176:5 through `price_m` → at examples/grid/station-leak:73:14 through `fetch_param` → declared as fetch_param, an op of Network declared at dsp/signal:66:1 at examples/grid/station-leak:427:1-430:2
```

This is what review cannot do: the build refuses, and names the line. Price belongs to the hour's plan, read before the reflex; a lookup inside the millisecond's decision is the bug.

It cannot reach the network while deciding, and the compiler proves it through every call. Every decision says why. A price read slipped into the reflex is refused with its path: decide → price_m → fetch_param → Network. The layer under Lasso.

# WHAT $5,000 DOES

a second ear and a GPS clock

Two ears are a stereo pair. The correlation meter's angle between two Salt Lake sites is the flow on the line between them. One ear here, one at GigaOne, and the mesh sees the line, not just the frequency.

I bring that measurement to Torus engineering.

### THE BUILD

github.com/ampactor-labs/hum
the ear, the director, the page, the data

### THE PROGRAM

github.com/ampactor-labs/mentl
examples/grid

### THE PAGE

ampactor.dev/hum
plays the recorded session, on a phone



THANK YOU, TORUS

# WHAT TORUS HAS, AND WHAT THIS ADDS

| TORUS TODAY, FROM ITS OWN PAGES | HUM |
| --- | --- |
| The Station monitors every cycle on three phases and corrects voltage in milliseconds (UL 1741 SB, IEEE 1547). | A room hears what the Station hears, from a guitar. Three estimators to the millihertz. Not a better sensor: a shared one. |
| Lasso forecasts, dispatches and learns, and reacts in milliseconds. | The program beneath a dispatcher: its negative capabilities proven, each decision explained in words, answered in under a millisecond. |
| Shield: a dedicated appliance, zero internet-facing attack surface. Overwatch: sub-second telemetry across the fleet. | A proof about the program itself, under the perimeter: it cannot reach the network while deciding, whatever the network does. Code an agent proposes passes the same gate or does not run. |
| Each Station answers frequency and voltage where it stands. | A suggestion with precedent: two ears with GPS time see the angle, which is the flow. FNET has done it from outlets since 2004. |



SOURCES: TORUS.CO, FNET/GRIDEYE

# THE GRID THROUGH A DSP LIBRARY'S LENS

| IN SONIDO | ON THE GRID |
|---|---|
| Pitch detection (the tuner, YIN) | Frequency, the one number the whole interconnection shares |
| THD (the distortion analyzer) | Power quality: the wave's timbre says what is plugged in (near a wall wart, mostly the wart's own saturation; the caption says so) |
| The Hilbert transform's instantaneous frequency | The textbook grid-frequency reading, sample by sample; its phase is the angle the mesh needs |
| A compressor | Droop control: threshold, ratio, ceiling, hold. IEEE 1547's frequency droop is a compressor on pitch |
| A noise gate | Islanding detection: the hum under its floor for six lines |
| Tuning to a reference, the beat going to zero | Reconnecting an island: a synchroscope is a tuner |
| The correlation meter (a stereo pair) | Two sites: the angle between them is the flow |
| LMS (the adaptive filter) | The learner, written twice: by hand in Rust, and derived by the compiler in Mentl |

# METHODS AND CALIBRATION

### CROSSINGS

Three bandpasses on the hum's harmonics, the loudest tracked. The period between rising zero crossings, interpolated, kept as a sample count plus a fraction so precision never decays. 100 ms and 1 s readings; the rate of fall over 300 ms.

### SPECTRUM

A two-second Hann window, zero-padded through sonido-analysis's FFT, the peak interpolated with a parabola on log magnitude. THD from its ThdAnalyzer on the same two seconds. The cross-check.

### LEARNED

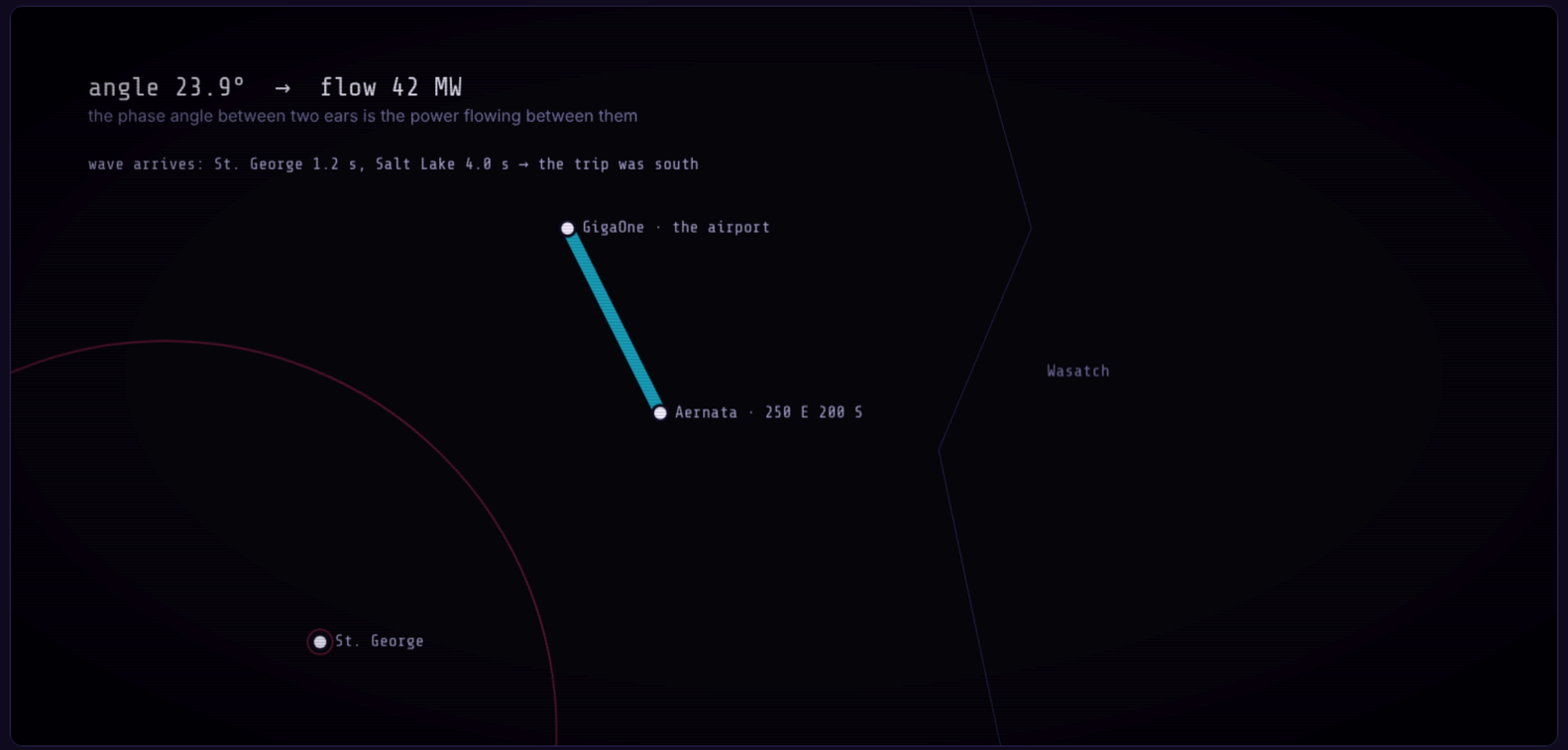
A one-coefficient predictor, $x[n] = a \cdot x[n-1] - x[n-2]$, behind a four-pole low-pass, its coefficient learned by a gradient the compiler derives from the chain itself. Within 5 mHz of the stream's own zero crossings. The third witness.

Measured on a synthesized hum with noise at −30 dB: the one-second reading within 0.5 mHz, the spectral reading within 0.2 mHz, the rate of change under 0.023 Hz/s at rest, a trip at 0.20 Hz/s detected 250 ms after it starts. The interface's clock offset is read from the hum's long-run mean and printed on the page.

# THE MESH

MESH

SIMULATED




Two Stations with GPS time see the **angle** between them, which is the flow on the line between them, and an event's wave arrives at one before the other. FNET has done this from wall outlets since 2004; a fleet that both sees and answers at 10C has not existed. Simulated from the day's shape and the replayed event.

# LATENCY AND SOURCES

Hum → ear: a line every 50 ms, the rate of fall over the last 300 ms → the director adds the hour, the rush window and the clean share → the controller answers on the same line, in under a millisecond → the page draws it a quarter second later. A trip at 0.20 Hz/s crosses the trigger 250 ms after it starts.

- EIA-930 hourly, PacifiCorp East: demand, day-ahead forecast, fuel mix. No key.
- NESO open data, one-second system frequency, 9 August 2019, the Hornsea and Little Barford trips, transposed to 60 Hz per unit.
- Torus's published figures for Spin and Pulse, from torus.co.
- FNET/GridEye, University of Tennessee, for the mesh's precedent.